University of Naples "Parthenope"

Department of Science and Technologies

M.Sc. in Applied Computer Science — Machine Learning & Big Data

Master's Thesis

Feature-Driven Bias Detection

An Association Rule Mining Approach to Analyze Feature Importance

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Section 1

Context and Motivation

Explainable Artificial Intelligence (XAI) and Fairness

- **XAI and the "Black-Box" Challenge**
  - High-performing models (e.g., Deep Learning) are often **black boxes** due to their inherent **opacity**.
  - XAI aims to make these decisions transparent, clarifying which factors influence a decision and how.
  - Essential in high-stakes contexts: **credit**, **healthcare**, **hiring**, and **justice**.

- **Bias and Fairness**
  - **The Bias Problem**: Systematic distortions or prejudices in data, collection procedures, or algorithms that lead to discriminatory outcomes.
  - **Fairness**: The ethical principle and normative requirement of ensuring that decisions are impartial and equitable across different individuals or groups.

Counterfactual Explanations and Feature Importance

- Given an instance x_i, its corresponding counterfactual x_i^{CF} is an instance belonging to the opposite class $f(x_i^{CF}) \neq f(x_i)$ whose difference with respect to x_i is minimal.
- Answers: *"What was the minimum change to x_i that would have led to a different decision?"*
- They are **local**, **model-agnostic** and **interpretable** explanations.
- According to recent literature, feature relevance can be measured based on counterfactuals: a feature is more relevant the more frequently its modification is required to flip the prediction (Alfeo et al., 2023).

Association Rule Mining (ARM) and Thesis Goal

- Born in **market basket analysis** (items bought together).
- A rule $A \Rightarrow B$ indicates that when A occurs, B is likely to occur too.
- **Our application**: items are the important features from counterfactuals. Extracted rules reveal the internal logic of the model.

Goal of the thesis

Use ARM to identify the main decision patterns of the model from relevant features obtained through counterfactuals. In particular, we focus on rules containing **biased** and **actionable** attributes.

Section 2

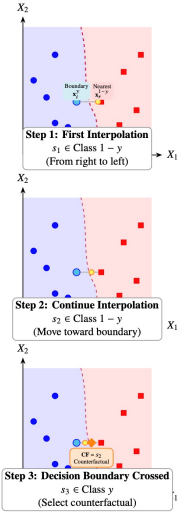
Methodology

Original BoCSoR Method (Alfeo et al., 2023)

- **BoCSoR**: *Boundary-based Counterfactual Scoring of Relevance* for tabular data.
- Designed for continuous spaces.

Procedure:

1. Compute for each x_i the closest instance of the opposite class.
2. Identify boundary instances: couples "instance–closest opposite instance" whose distance is less than a fixed percentile p.
3. Compute 5 synthetic instances for each couple via linear interpolation, identifying as counterfactual x_i^{CF} for x_i the closest synthetic instance belonging to the opposite class.
4. Perturbation step: substitute features x_{ij} in x_i^{CF} to check for a class change after re-classification, recording feature j as relevant if the class changes.
5. Compute BoCSoR score as global feature importance metric for each feature j.



Issues of BoCSoR for Categorical Data

Most real-world datasets contain tabular **categorical data**, especially in census data, where even quantitative numerical columns such as *age* and *hours worked* are organized into bands.

The problems of adapting BoCSoR to categorical data are:

- Usage of Euclidean distance
- Interpolation step

In particular, in categorical cases features may be **categorical ordinal** or **categorical nominal**.

AGEP	COW	SEX	SCHL	WKHP	...
Young	Emp.-Private-For-Profit	Female	Bachelors-Degree	Over-Full-time	...
Young-Adult	Emp.-Private-For-Profit	Female	Regular-HS-Diploma	Part-time	...
Retirement-Age	Local-Government-Employee	Male	Masters-Degree	Full-time	...
...	...	...	...	...	...

Hybrid Encoding and Unified Distance

- Before explaining BoCSoR adaptation to categorical case, it is necessary to define how nominal and ordinal data are encoded and processed.
- Categorical space: $\tilde{x}_i = (x_i^{nom}, x_i^{ord}) \in \tilde{\mathcal{X}}$
- Each feature X_j is encoded to represent each possible categorical value assigned to it $c_{jk} \in C_j$, introducing new features for each instance: $enc(\tilde{x}_{ij}) = c_{jk}$
- **Encoding strategy**:
  - **Nominal**: *one-hot encoding* (scaled by 0.5) in $\{0, 1\}$.
  - **Ordinal**: *rank encoding* + *min-max scaling* in $[0, 1]$.
- **A single distance**: the distance between two instances is the global Manhattan distance, defined as the sum of the Manhattan distances across the individual encoded features, ensuring coherence between heterogeneous types:

$$d_{man}(\tilde{x}_a, \tilde{x}_b) = \sum_j d_{man}\big(enc(\tilde{x}_{aj}),\ enc(\tilde{x}_{bj})\big)$$

Examples of Ordinal and Nominal Encoding

Example (ordinal only). Consider two instances $\tilde{\mathbf{x}}_a$ and $\tilde{\mathbf{x}}_b$ on the same j-th ordinal feature, with four ordered categories $(c_{j1} \prec c_{j2} \prec c_{j3} \prec c_{j4})$ and $p_j = 4$. Suppose that $x_{aj}^{\text{ord}} = c_{j2}$ and $x_{bj}^{\text{ord}} = c_{j4}$.

$$\text{rank}(x_{aj}^{\text{ord}}) = 2, \qquad \text{rank}(x_{bj}^{\text{ord}}) = 4$$

After min-max normalization,

$$\text{ord_enc}(x_{aj}^{\text{ord}}) = \frac{2-1}{4-1} = \frac{1}{3}, \qquad \text{ord_enc}(x_{bj}^{\text{ord}}) = \frac{4-1}{4-1} = 1$$

The Manhattan distance on this ordinal feature is

$$d_{\text{Man}}\Big(\text{ord_enc}(x_{aj}^{\text{ord}}), \text{ord_enc}(x_{bj}^{\text{ord}})\Big) = \left|\frac{1}{3} - 1\right| = \frac{2}{3}$$

Hence, each ordinal-feature contribution lies in $[0, 1]$.

Example (nominal only). Let $C_j = \{c_1, c_2, c_3\}$ and consider two instances $\tilde{\mathbf{x}}_a$ and $\tilde{\mathbf{x}}_b$ such that $x_{aj}^{\text{nom}} = c_1$ and $x_{bj}^{\text{nom}} = c_3$.

$$\text{one_hot}(x_{aj}^{\text{nom}}) = (1, 0, 0), \qquad \text{one_hot}(x_{bj}^{\text{nom}}) = (0, 0, 1)$$

The Manhattan distance on this unscaled block is

$$d_{\text{Man}}\Big(\text{one_hot}(x_{aj}^{\text{nom}}), \text{one_hot}(x_{bj}^{\text{nom}})\Big) = 2$$

After scaling by $1/2$,

$$\text{nom_enc}(x_{aj}^{\text{nom}}) = (0.5, 0, 0), \qquad \text{nom_enc}(x_{bj}^{\text{nom}}) = (0, 0, 0.5)$$

Evaluating the Manhattan distance on these scaled encodings yields

$$d_{\text{Man}}\Big(\text{nom_enc}(x_{aj}^{\text{nom}}), \text{nom_enc}(x_{bj}^{\text{nom}})\Big) = 1$$

Hence, each nominal-feature distance contribution is within $\{0, 1\}$.

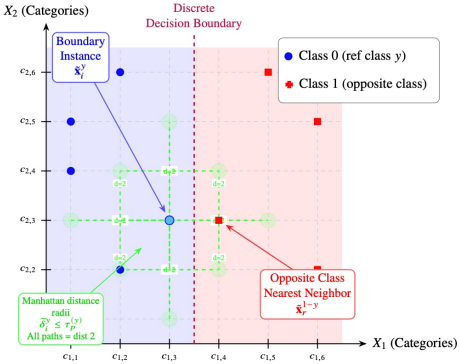
Adaptation to the Categorical Case

The space $\tilde{\mathcal{X}}$ is **categorical**:

- Euclidean $\longrightarrow$ **Global Manhattan** distance.
- Linear interpolation $\longrightarrow$ **k-nearest opposite-class neighbors** for each categorical instance $\tilde{x}_i$, since in a categorical space perturbing the closest counterfactual cannot be enough to flip the class label.
- Perturbation step records a list of relevant feature = category pair $X_j = c_{jk}$ for each instance across all k-counterfactuals:

$$L_i = \bigcup_{r=1}^{k} \left\{ (X_j = c_{jk}) \,\middle|\, j \in \{1, \ldots, n_{cat}\},\ f(\tilde{x}_{i,r}^{CF,j}) = f(\tilde{x}_i) \right\}$$

- Store $\mathcal{E}$ the set of all L_i for each $\tilde{x}_i$, considered as a transactional database.



ARM in the Thesis Context

- **Input Data**: ARM operates on the lists of relevant features (L_i) extracted in the previous step.
- **Feature Space**: these lists belong to the transaction database $\mathcal{E}$.
- **Transaction Representation**: each instance is modeled as a transaction of its specific relevant features.

Classic ARM (Market Basket)

List of items: {Milk, Bread, Butter, . . . }

x_1 = {Milk, Cereal, Butter}
x_2 = {Bread, Milk}
x_3 = {Butter, Eggs, Flour}

Our Proposed Case

Items = relevant feature–category pairs extracted from counterfactuals.

$x_1 = \{f_3, f_6, f_8\}$
$x_2 = \{f_1, f_3\}$
$x_3 = \{f_2, f_8, f_{10}\}$

Example Rule Comparison:

- **Classic Market Basket**: $\{\text{Butter}\} \Rightarrow \{\text{Milk}\}$
- **Feature-Based Case**: $\{f_1\} \Rightarrow \{f_3\}$

ARM Metrics: Evaluating Feature Associations

Goal: Quantifying the strength and reliability of the extracted rules $X \Rightarrow Y$ (e.g., $f_1 \Rightarrow f_3$). Let N be the total number of samples.

- **Support**: Measures the frequency of the pattern. How often do features X and Y appear together as relevant in the dataset?

$$\text{Support}(X \Rightarrow Y) = \frac{\text{freq}(X \cup Y)}{N}$$

- **Confidence**: Measures the reliability of the inference (conditional probability). If feature X is relevant, how likely is it that Y is also relevant?

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

- **Lift**: Measures the strenght of correlation of the association compared to random chance.

$$\text{Lift}(X \Rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

Lift > 1: positive association Lift $= 1$: independence Lift < 1: negative association.

Hierarchical ARM: Why Two Levels?

Applying ARM directly to $\mathcal{E}$ raises two issues:

- **Combinatorial explosion**: with n_{cat} features and up to p_j categories each, the rule space grows as $\left(\sum_j p_j\right)^k$ for itemsets of size k.
- **Interpretive fragmentation**: rules like {SCHL=Bachelors} $\Rightarrow$ {OCCP=Tech} and {SCHL=Masters} $\Rightarrow$ {OCCP=Prof} are reported as independent, yet both instantiate the same coarser pattern: SCHL and OCCP tend to be jointly relevant.

Solution: organize the analysis into *two levels of granularity*.

Macroscopic level — *features only*:
which features are jointly relevant?
Example: {SCHL} $\Rightarrow$ {OCCP}

Microscopic level — *feature = category pairs*:
which categorical combinations realize each macro rule?
Example: {SCHL=Bach.} $\Rightarrow$ {OCCP=Tech}

Macroscopic Analysis

Projection onto feature names. For each local explanation $\mathcal{L}_i \in \mathcal{E}$, keep only the feature indices, discarding the categories:

$$\mathcal{M}_i = \{\, X_j \mid (X_j = c_{jk}) \in \mathcal{L}_i \,\}, \qquad \mathcal{M} = \{\mathcal{M}_i \mid \mathcal{L}_i \in \mathcal{E}\}.$$

Mining. Run ARM on $\mathcal{M}$ to obtain candidate rules $A_{\text{macro}} \Rightarrow B_{\text{macro}}$ between *feature names*.

Filtering thresholds (macroscopic)

A candidate rule is kept in $\mathcal{R}_{\text{macro}}$ if and only if:

- $\text{supp}(R) \geq \tau_{\text{supp}}^{\text{macro}}$ (frequent across boundary instances)
- $\text{conf}(R) \geq \tau_{\text{conf}}^{\text{macro}}$ (reliable implication)
- $\text{lift}(R) \notin [1-\epsilon,\ 1+\epsilon]$ (non-independence: discard chance co-occurrence)

Output. The set $\mathcal{R}_{\text{macro}}$ of *structural dependencies* between features, e.g.:
$\{\text{SCHL}\} \Rightarrow \{\text{OCCP}\}, \quad \{\text{SCHL}, \text{AGEP}\} \Rightarrow \{\text{OCCP}\}.$

Microscopic Analysis

Per-rule restriction. For each macro rule $R \in \mathcal{R}_{\text{macro}}$, consider only the transactions in $\mathcal{E}$ that contain a category for *every* feature of R:

$$\mathcal{E}_R = \{\, \mathcal{L}_i \in \mathcal{E} \mid \forall X_j \in A_{\text{macro}} \cup B_{\text{macro}},\ \exists\, (X_j = c_{jk}) \in \mathcal{L}_i \,\}.$$

Running ARM on $\mathcal{E}_R$ yields rules $A_{\text{micro}} \Rightarrow B_{\text{micro}}$ on feature = category pairs, each being a specialization of R.

Filtering thresholds (microscopic)

Within $\mathcal{E}_R$, support is bounded by $|\mathcal{E}_R| \ll |\mathcal{E}|$ and is split across categories: lower thresholds are required.

- $\text{supp}(r) \geq \tau_{\text{supp}}^{\text{micro}}$, with $\tau_{\text{supp}}^{\text{micro}} < \tau_{\text{supp}}^{\text{macro}}$
- $\text{conf}(r) \geq \tau_{\text{conf}}^{\text{micro}}$, with $\tau_{\text{conf}}^{\text{micro}} < \tau_{\text{conf}}^{\text{macro}}$
- $\text{lift}(r) \notin [1-\epsilon,\ 1+\epsilon]$ (same independence window)

Output. A mapping $\mathcal{R}_{\text{micro}} : R \mapsto \mathcal{R}_{\text{micro}}(R)$ that attaches to each macro rule its category-level specializations. Example for {SCHL} $\Rightarrow$ {OCCP}:
{SCHL=Bachelors} $\Rightarrow$ {OCCP=Tech-support}.

Section 3

Dataset and Data Preparation

Limitations of Classic Tabular Benchmarks

- In Fairness/XAI literature, recurring benchmarks like **Adult**, **COMPAS**, and **German Credit** are standard.
- While useful for methodological comparisons, they have **limited empirical coverage**: small sample sizes, dated historical contexts, and low heterogeneity.
- They often exhibit **weak anonymization**, **hidden proxies**, and **weak documentation**, leading to reduced semantic richness.
- Issues like **missing values**, **encoding artifacts**, and **preprocessing errors** can bias both model performance and fairness assessments.

From UCI Adults to Folktables

The Adult problem (Ding et al., 2021), standard de facto for census data:

- Data from the 1994 CPS; the fixed $50,000 threshold is no longer socio-economically meaningful.
- $< 50k$ records · 14 features · no geographic stratification.
- Coding artifacts (`fnlwgt`, a demographic weight) often misused as predictive features.
- Weak anonymization and hidden proxies limit semantic richness.

The solution: Folktables (Ding et al., 2021)

- Recent ACS PUMS microdata from the US Census Bureau.
- Reproducible and configurable by year, task and geography.
- Focus task: **ACSIncome** (binary income classification).
- Key advantage: bias varies across socio-economic contexts.

Data Preparation and Augmentation

Benchmark	Size	Records	Threshold θ	Socio-Economic Profile
NY Metropolitan Area	Small	106,877	\$94,400	Urbanized and rich
Texas	Small	145,670	\$90,800	Poor and rural
Northeast	Medium	508,138	\$100,700	Urbanized and rich
South	Medium	642,617	\$86,000	Poor and rural
ALL (U.S. except AK)	Large	1,753,049	\$94,200	Highly heterogeneous (National baseline)

Section 4

Results

Experimental Setup and Semantic Classification

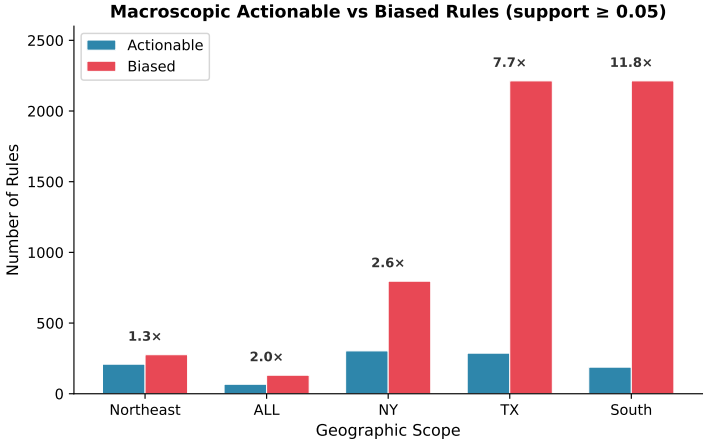
Setup (320 configurations)

- **Data source**: ACS **2024**.
- **5 geographic benchmarks**: NY, TX, Northeast, South, ALL.
- **2 models**: CatBoost (axis-aligned splits), MLP (smooth surfaces).
- **8 neighborhood sizes**: $k \in \{1, 3, 5, 7, 9, 11, 13, 15\}$.
- **4 boundary percentiles**: $P \in \{10, 15, 20, 25\}\%$.
- **Rule thresholds**: different values of support and confidence were evaluated via grid search, while a fixed lift window width of **25** was used.

Rule Classification

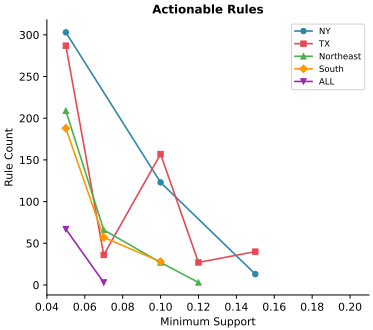
- **Actionable**: only modifiable features (SCHL, COW, WKHP, OCCP).
- **Biased**: contains at least one sensitive feature (SEX, RAC1P).
- **Other**: contextual (AGEP, MAR, RELP, POBP).

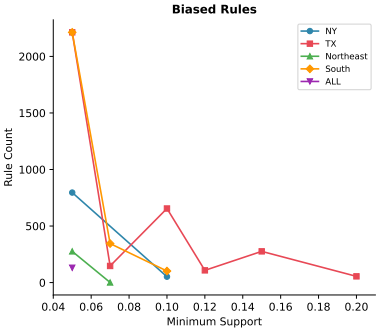
Macroscopic Results: Geographic Asymmetry



- Dominant **actionable**: COW ⇒ OCCP.
- Dominant **biased**: RAC1P ⇒ OCCP.
- **South**: unique pattern RAC1P ⇒ POBP (Race–Birthplace proxy).
- Biased/Actionable ratio: **11.8**× in the South vs **1.3**× in NE.

Decay Analysis: Which Rules Persist?





- **NE & NY**: actionable rules persist at higher support levels.
- **TX & South**: biased rules are extremely robust (up to supp ≥ 0.20).
- **National data (ALL)** dilutes local bias patterns.

Microscopic Results: Profiling the Bias

- **Texas – Gender bias in education:**
  $\{\text{COW=Local-Gov},\ \text{SEX=Female}\} \Rightarrow \{\text{OCCP=Education}\}$
- **New York – Occupational segmentation:**
  $\{\text{RAC1P=White},\ \text{SEX=Female}\} \Rightarrow \{\text{OCCP=Office-Admin}\}$
- **South – Geography as racial proxy:**
  $\{\text{RAC1P=Black}\} \Rightarrow \{\text{POBP=Georgia}\}$

Section 5

Conclusions

Conclusions and Future Work

Main Findings

- ARM + Counterfactuals is effective for global XAI and bias detection.
- Identified clear geographic asymmetries in ML model prediction.
- Hierarchical design is key to understanding the “how” and “why” of decison pattern.

Future Work

- Temporal analysis (fairness drift).
- Integration with Causal Inference.
- Evaluation on other types of classifiers.

Thank you for your attention

Alfredo Mungari

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